TARLOGIC
CYBERSECURITY EXPERTS

ROOTEDCON MADRID 2026

HACKING BLUETOOTH AT SCALE

SPEAKERS

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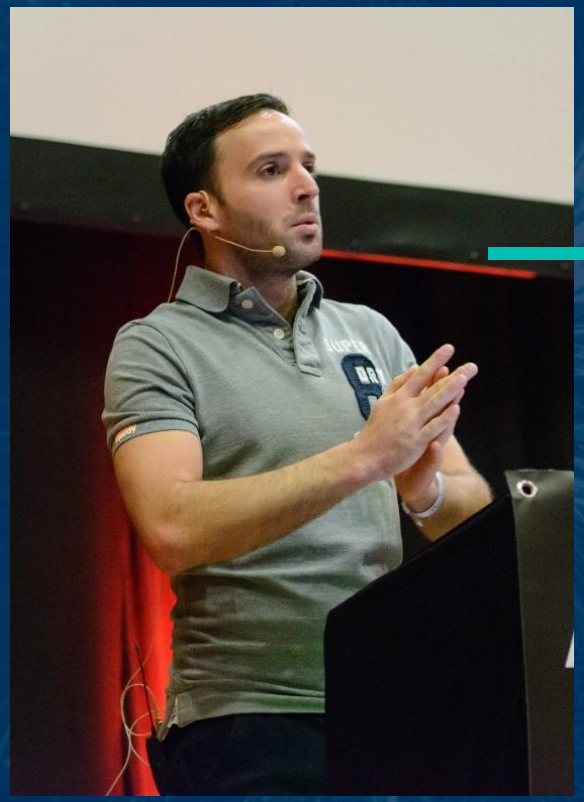

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01 INTRODUCTION

INTRODUCTION

2021

Need to perform Bluetooth audits, but the information is scattered and incomplete.

Research for an in-depth understanding of Bluetooth technology

2022

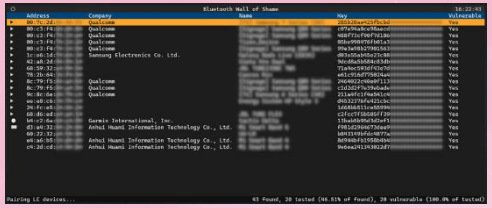
BlueTrust
Cloning Bluetooth identities and identifying trust relationships between devices

Development of comprehensive and accesible documentation

2023

BSAM
Complete, public and open-source Bluetooth auditing methodology

INTRODUCTION



2024

BlueSpy & WallOfShame
Proof of concept to listen to audio from headphones

Accessibility to tools

2025

UsbBluetooth & ESP32 commands
Libraries for developing tools

C/C++ .net

Need for auditing tools

2026

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02 BSAM

BSAM

BLERP: BLE Re-Pairing Attacks and Defenses

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***Abstract*—Bluetooth Low Energy (BLE) is a ubiquitous wireless technology used by billions of devices to exchange sensitive data. As defined in the Bluetooth Core Specification v6.1, BLE security relies on two primary protocols: *pairing*, which establishes a long-term key, and *session establishment*, which encrypts communications using a fresh session key. While the standard permits paired devices to *re-pair* to negotiate a new security level, the security implications of this mechanism remain unexplored, despite the associated risks of device impersonation and Machine-in-the-Middle (MitM) attacks.**

We analyze BLE re-pairing as defined in the standard v6.1 and identify six design vulnerabilities, including four novel ones, such as unauthenticated re-pairing and security level downgrade. These vulnerabilities are design flaws and affect any standard-compliant BLE device that uses pairing, regardless of its Bluetooth version or security level. We also present four new re-pairing attacks exploiting these vulnerabilities, which we call BLERP. The attacks enable impersonation and MitM of paired devices with minimal or no user interaction (1-click or 0-click). Our attacks are the first to target BLE re-pairing, exploit the interplay between BLE pairing and session establishment, and abuse the SMP security request message.

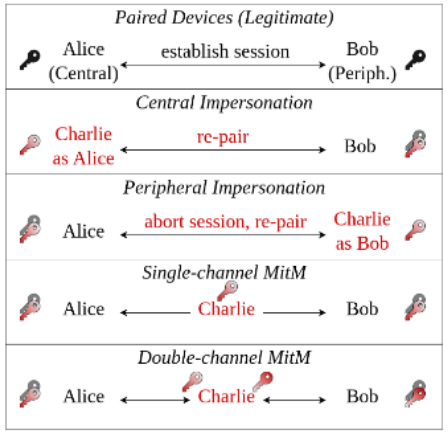


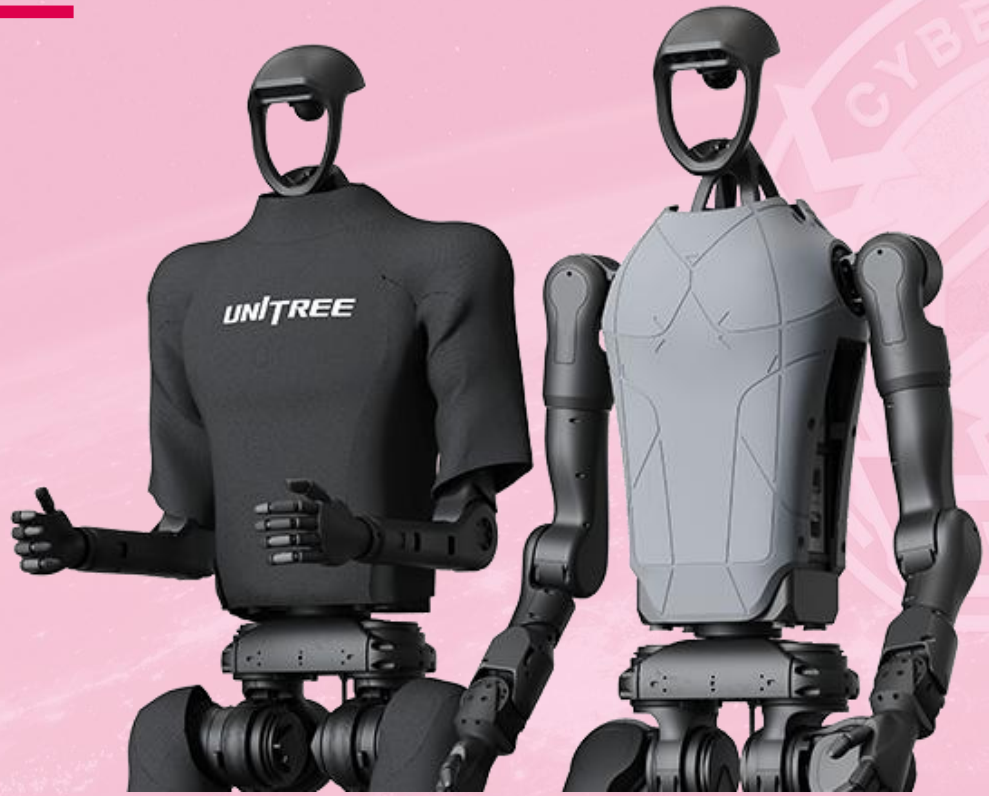
Fig. 1: BLERP attacks. Alice and Bob have paired over BLE. Charlie can perform Central and Peripheral impersonation

Published in February 2026

https://github.com/sacca97/blerp/

- BSAM-PA-04 - **Rejection of legacy pairing**
- BSAM-PA-05 - **Pairing without user interaction**
- BSAM-PA-08 - **Bluetooth link key removal**

BSAM



Published in September 2025

https://github.com/bin4ry/UniPwn

- BSAM-SE-03 - **Bluetooth Service access control**
- BSAM-AP-07 - **Secure services**

BSAM

Three years in use on very diverse devices…

- Medical devices, cryptographic devices, inventory and logistics systems, pet accessories…

Highly valued and requested by manufacturers!

It is not trivial to perform an evaluation and tools are still missing…

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03 AUTOMATION

AUTOMATION

BSAM consists on:

- 7 categories
- 36 controls

16 controls can be automated…

Discovery	5
Pairing	7
Encryption	2
Services	2

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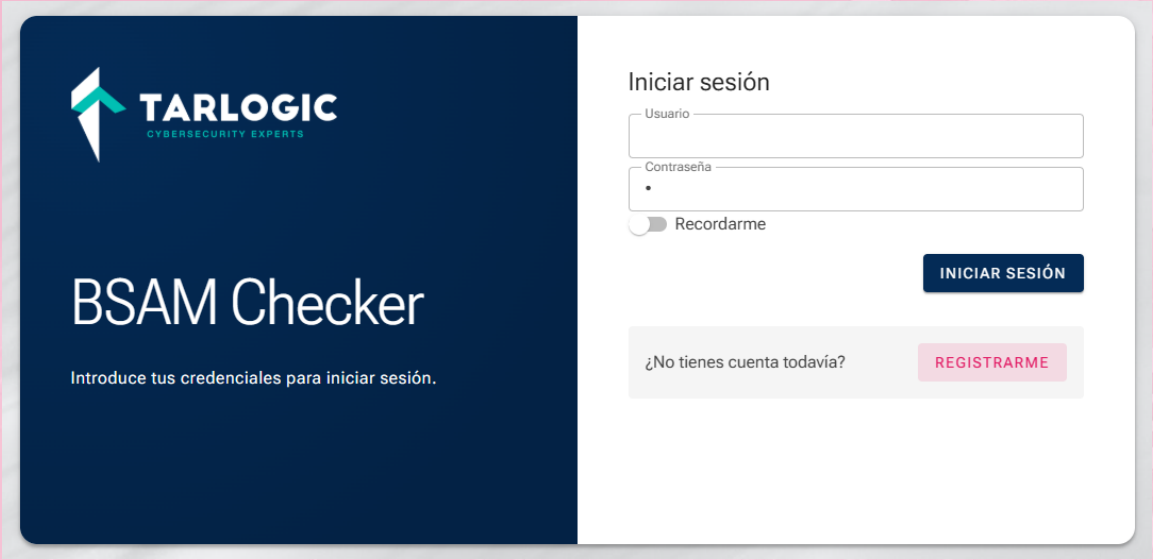
04 BSAM CHECKER

BSAM CHECKER



Desktop application

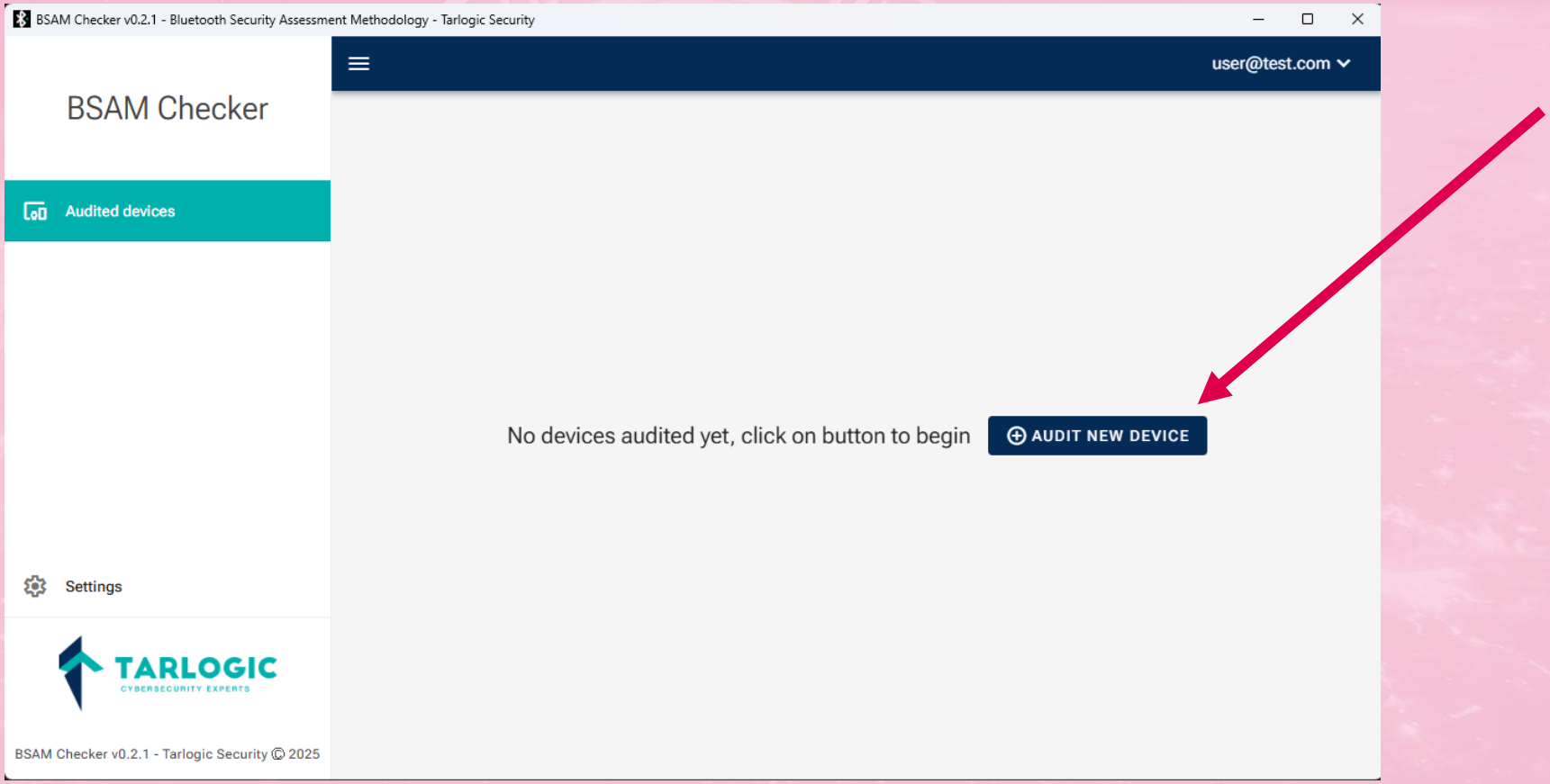
- Requires a *Bluetooth dongle*

Web version

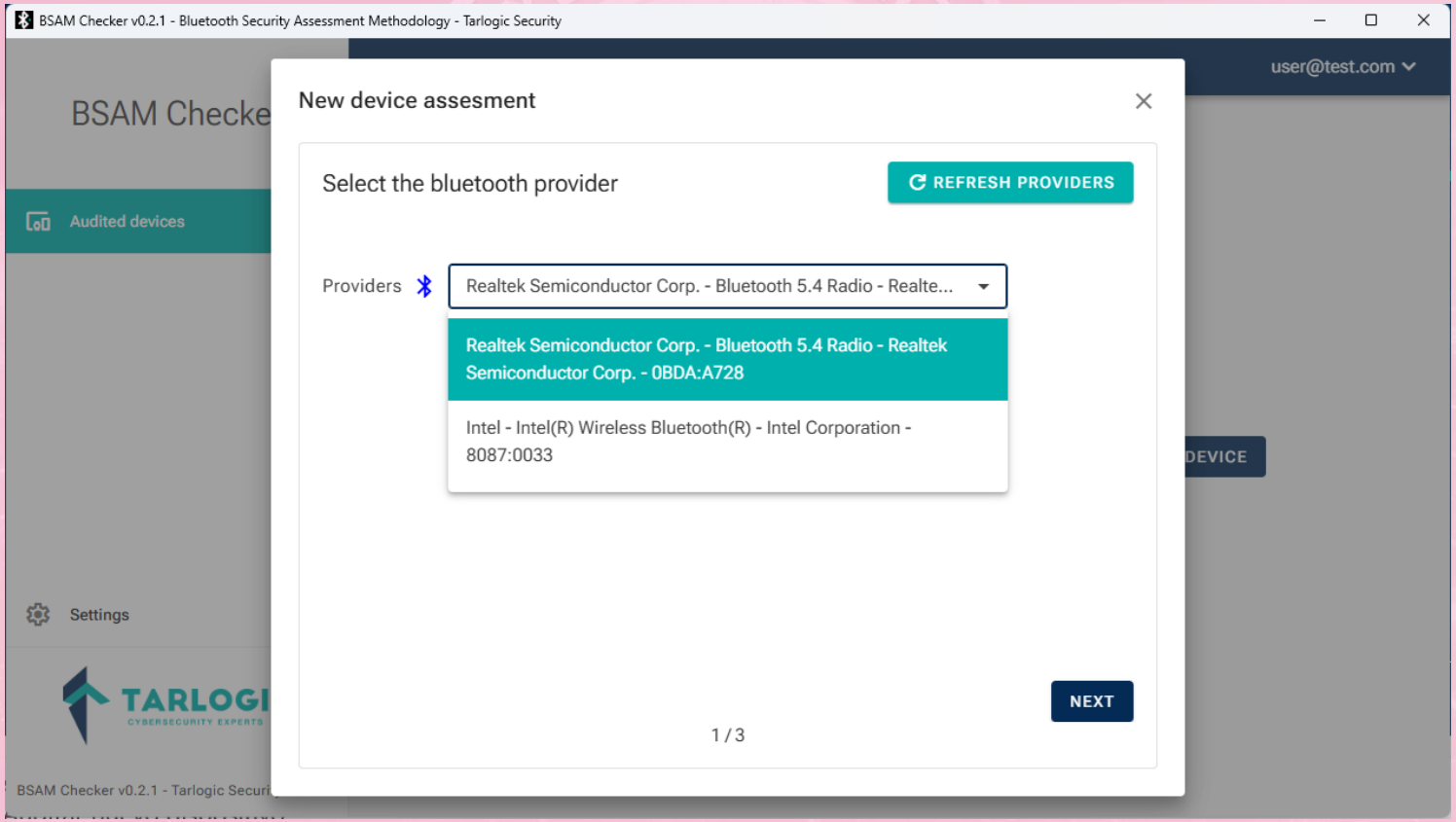
- To Access the results repository

Completely free tool!

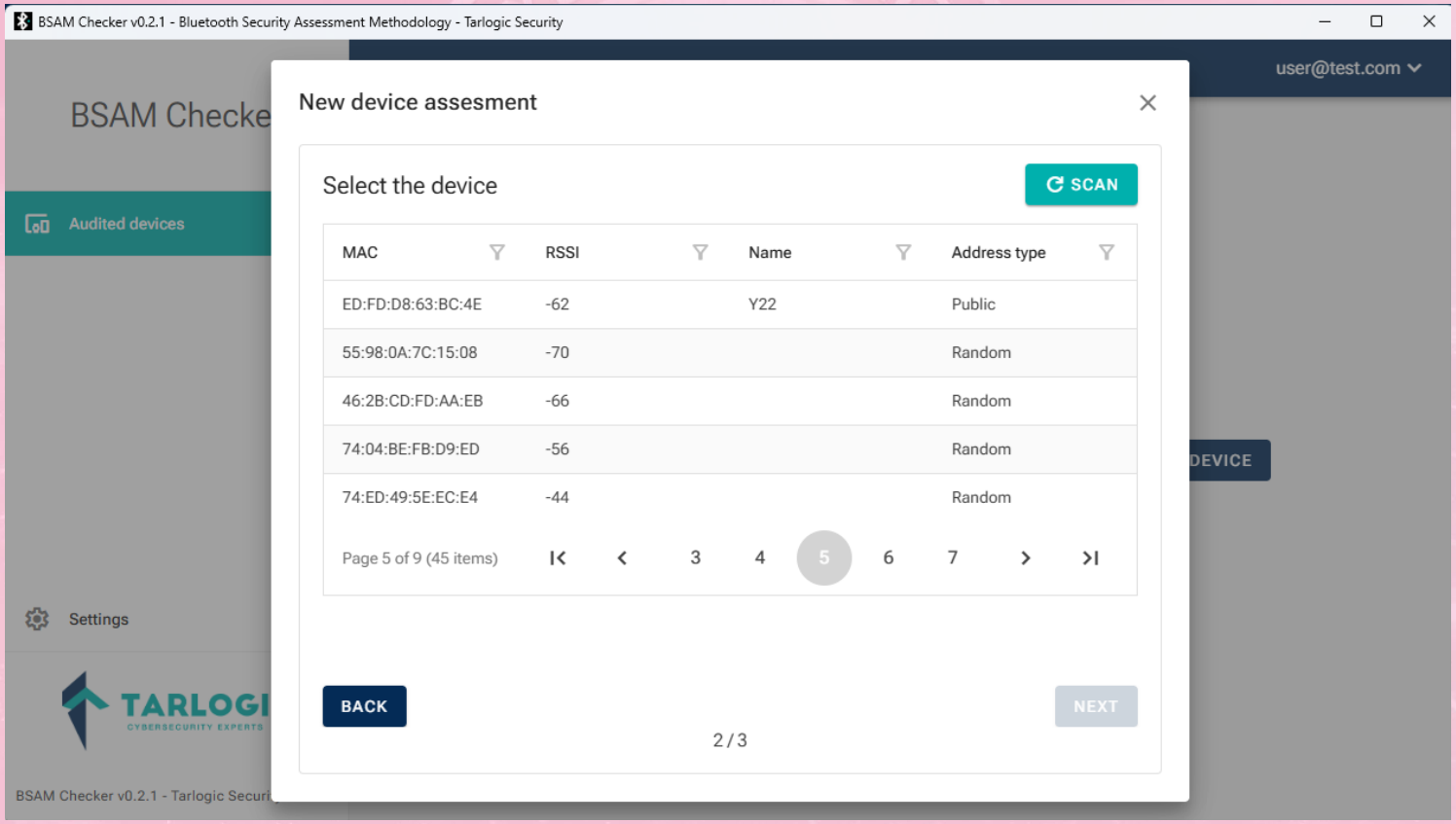
BSAM CHECKER



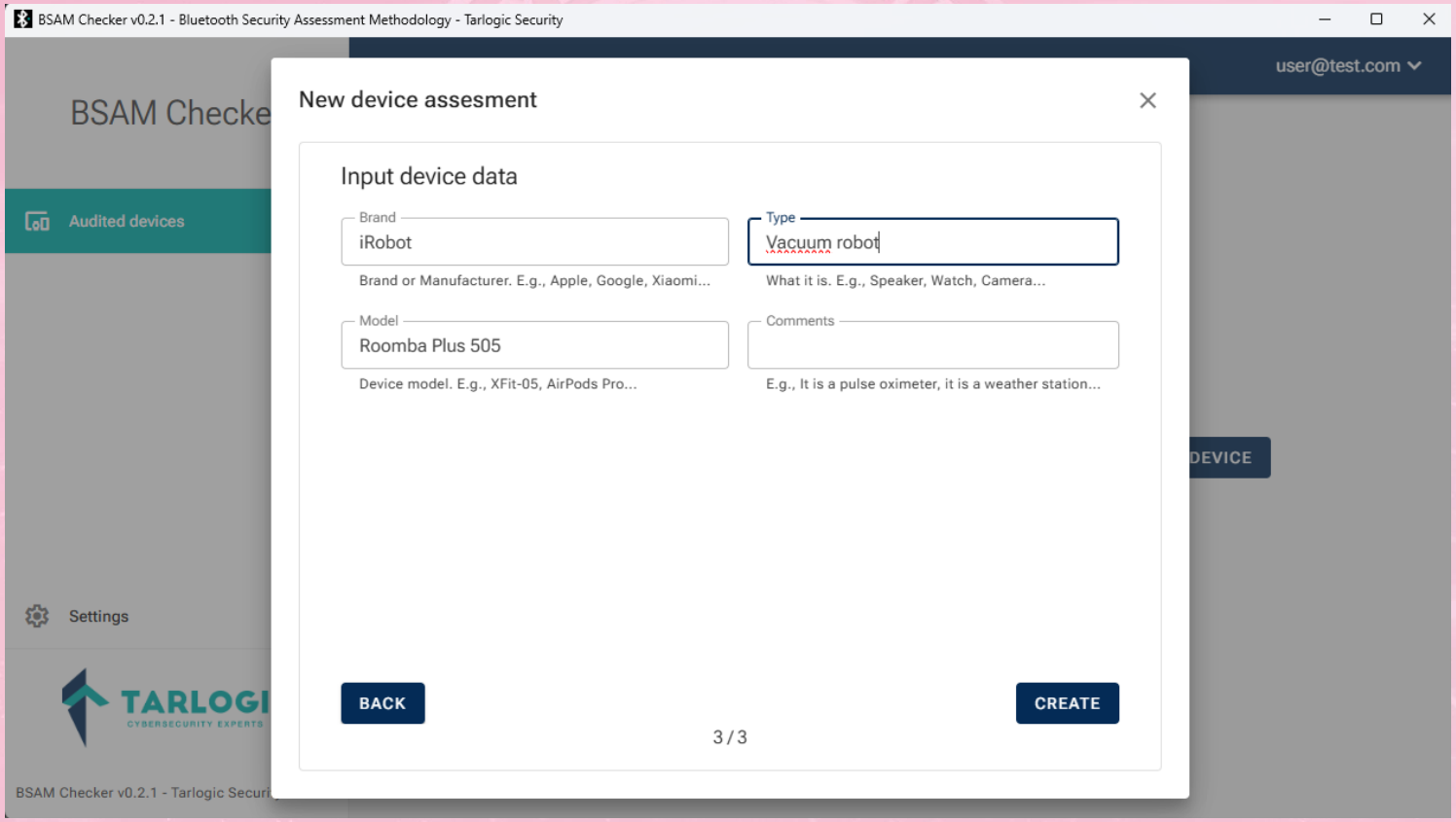
BSAM CHECKER



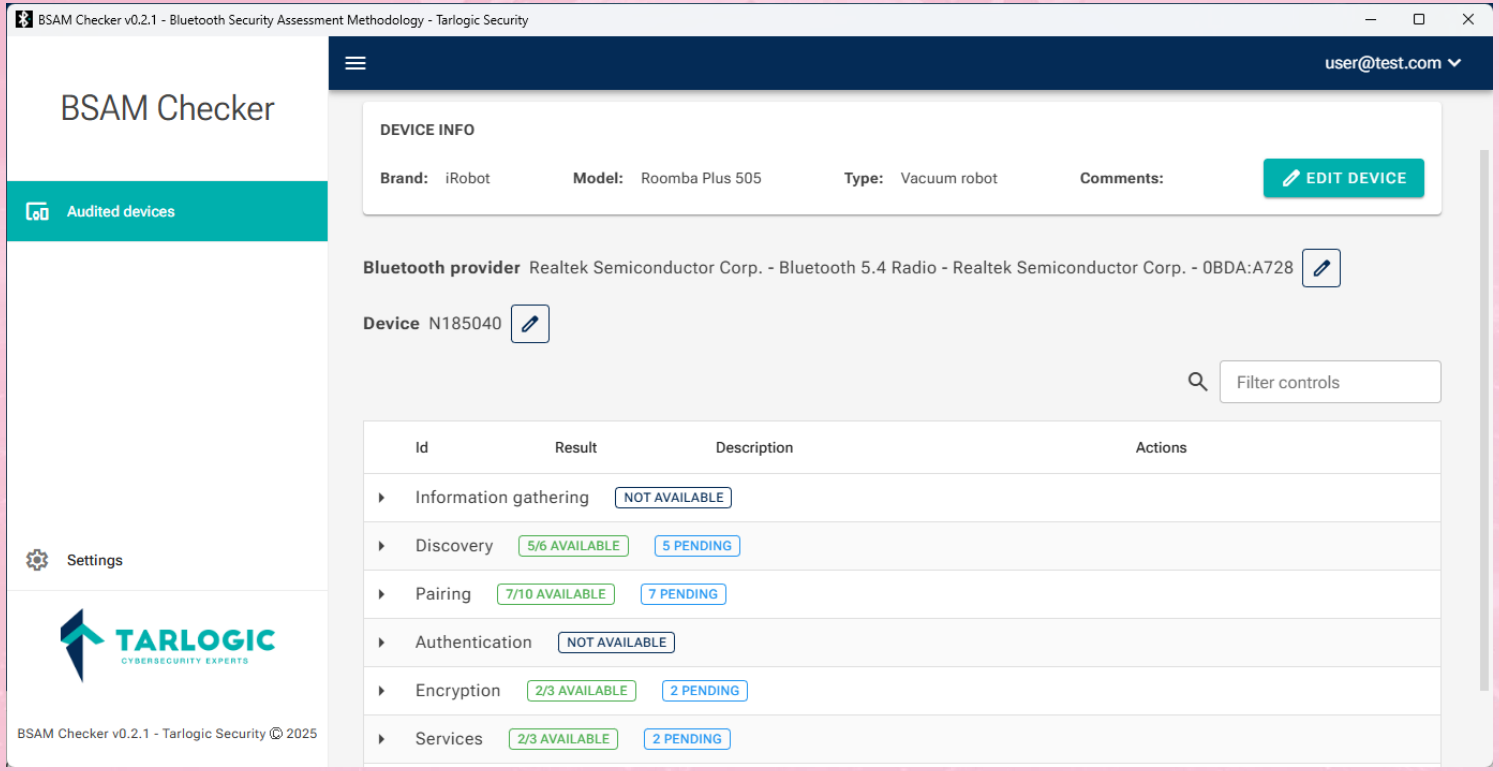
BSAM CHECKER



BSAM CHECKER



BSAM CHECKER



BSAM CHECKER

A control has 4 posible results:

Vulnerable

The vulnerability is present in the device and can be exploited.

Not vulnerable

The device is secure for that control.

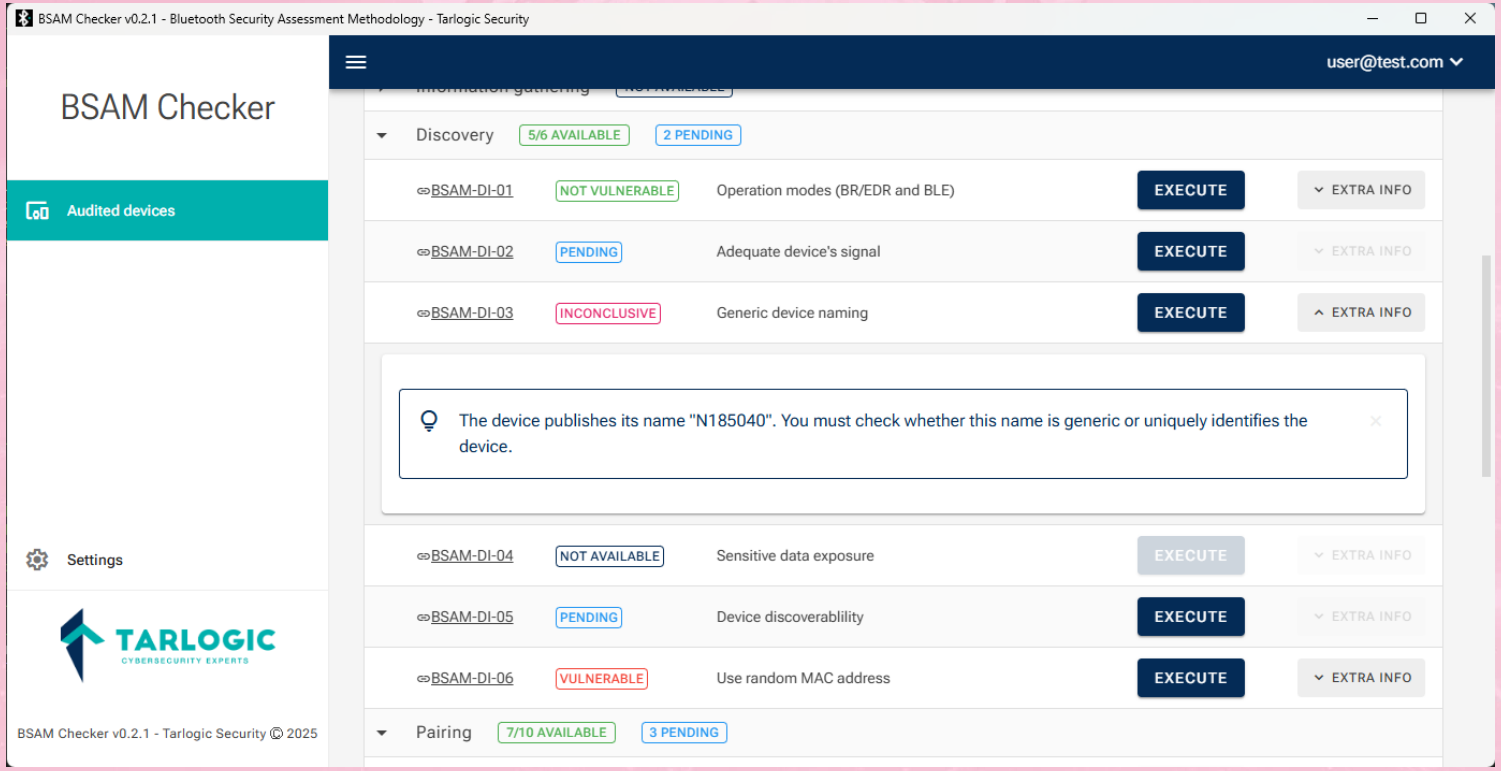
Inconclusive

Evidence was found that does not allow determining whether it is vulnerable or not automatically and the user must review the results to determine it manually.

Error

The execution control failed; the details include information about the error and steps to rerun and avoid the problem.

BSAM CHECKER



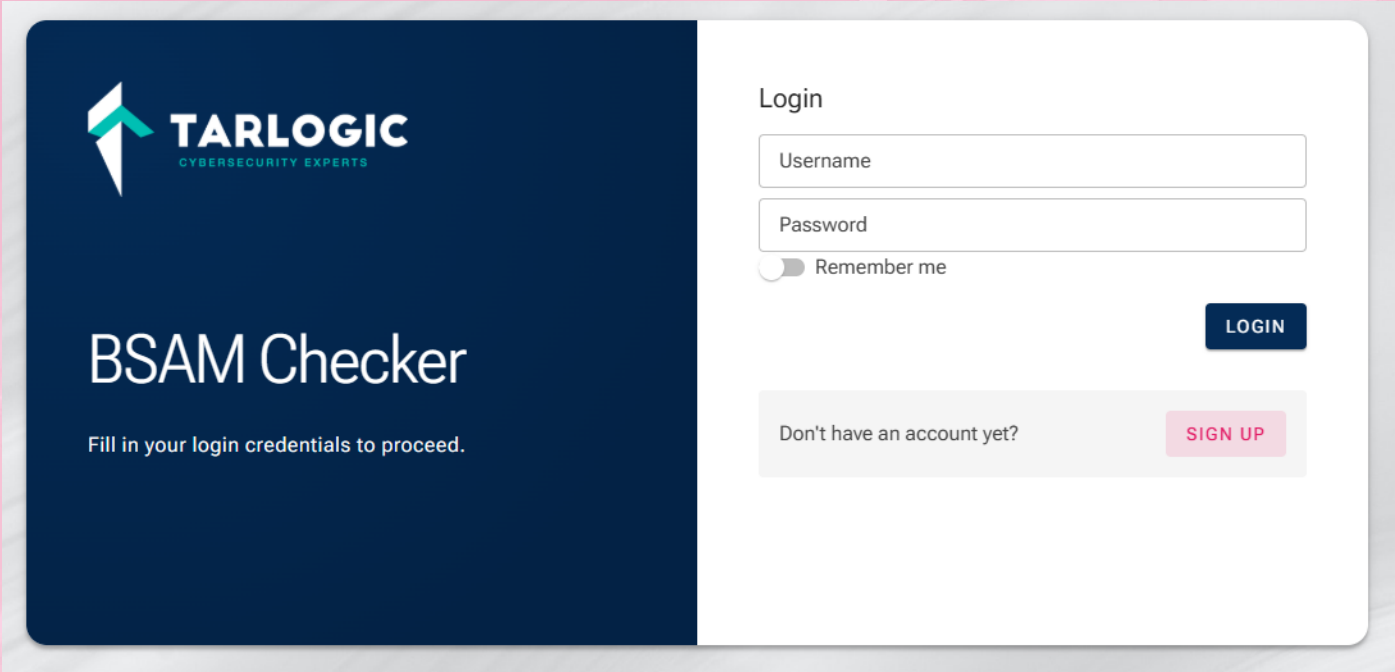
BSAM CHECKER

```
 _____ _____ _____ _____    _____ _____ _____ _____ _____ _____ _____
| __  |   __|  _  |     |  |     |  |  |   __|     |  |  |   __| __  |
| __ -|__   |     | | | |  |  --|     |   __|   --|    -|   __|    -|
|_____|_____|__|__|_|_|_|  |_____|__|__|_____|_____|__|__|_____|__|__|
 Bluetooth Security Assessment Methodology - Tarlogic Security -  v0.2.1
                                                  27/02/2026 21:01:12


Información del dispositivo
Marca               : Samsung
Modelo              : QN65Q800TAFXZA
Tipo                : TV
Comentarios         : bedroom tv
Fecha de creación   : 17/12/2025 12:57:10


Resultados de los controles:
 [No disponible  ] | BSAM-IG-01 | Ciclo de vida del controlador Bluetooth
 [No disponible  ] | BSAM-IG-02 | Vulnerabilidades en el controlador Bluetooth
 [No disponible  ] | BSAM-IG-03 | Vulnerabilidades de la pila Bluetooth del host
 [No disponible  ] | BSAM-IG-04 | Vulnerabilidades conocidas en estándar Bluetooth
 [No concluyente ] | BSAM-DI-01 | Modos de operación (BR/EDR y BLE) | El dispositivo afirma ser compatible con Bluetooth BR/EDR y LE.
                                  Evalúa si realmente se necesitan ambos modos o si el dispositivo podría funcionar solo con uno de ellos.
 [Pendiente      ] | BSAM-DI-02 | Potencia de señal adecuada
 [No concluyente ] | BSAM-DI-03 | Nombre del dispositivo genérico .
                                  Comprueba si este nombre es genérico o identifica de forma única al dispositivo.
 [No disponible  ] | BSAM-DI-04 | Exposición de datos sensibles
 [Vulnerable     ] | BSAM-DI-05 | Descubrimiento de dispositivo | El dispositivo está anunciado. Esto significa que cualquiera puede verlo.
 [No vulnerable  ] | BSAM-DI-06 | Uso de direcciones MAC aleatorias | El dispositivo anuncia una dirección MAC marcada como aleatoria. Valida
 [Pendiente      ] | BSAM-PA-01 | Modo de emparejamiento por defecto
```

DEMO (1)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BSAM Checker
Fill in your login credentials to proceed.
Login
Username
Password
Remember me
LOGIN
Don't have an account yet?
SIGN UP

BSAM Checker v0.2.1 - Bluetooth Security Assessment Methodology - Tarlogic Security

BSAM Checker

Audited devices

No devices audited yet, click on button to begin

AUDIT NEW DEVICE

Settings

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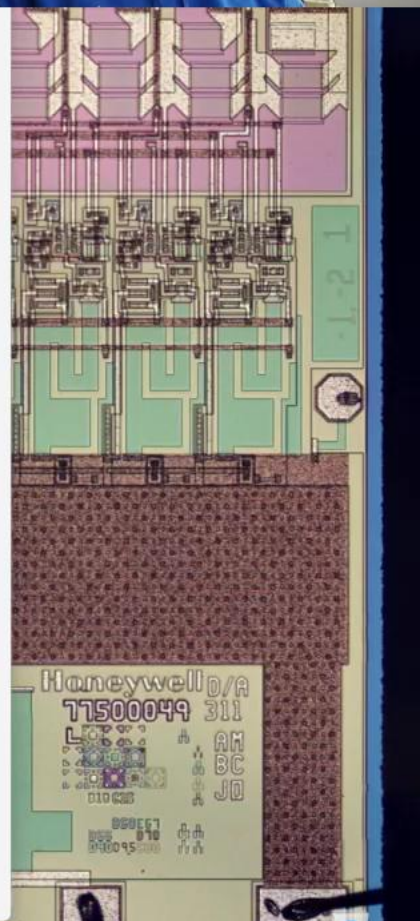
BSAM Checker v0.2.1 - Tarlogic Security © 2026



Papelera de reciclaje

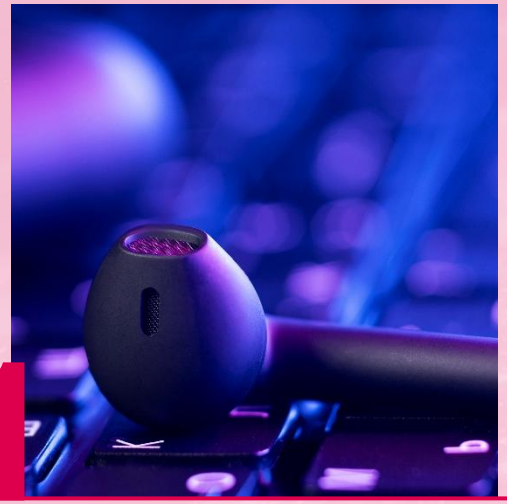
C:\pentest\Tarlogic\bsa

BSAM Checker v0.2.1 - B

ESP

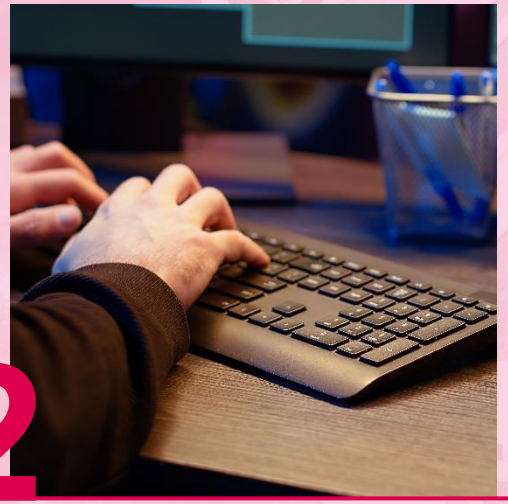
17:36
03/03/2026

DEMO (1)



1

- The device is **vulnerable** to all pairing controls...



2

- This allows unrestricted Access to all services…

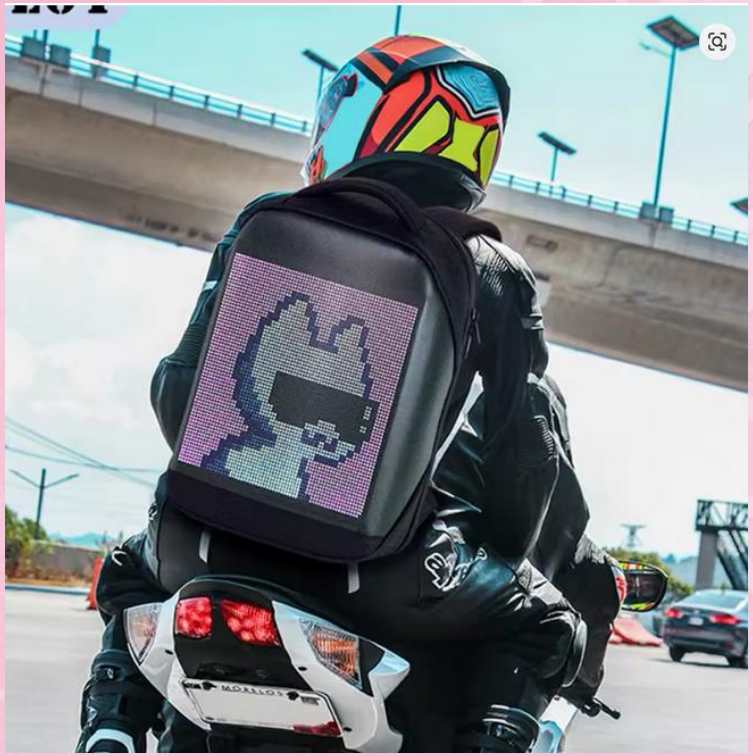
3

- https://github.com/TarlogicSecurity/BlueSpy

4

- **Samsung:** CVE-2024-58101 (CVSSv4.0 8.5 HIGH)

DEMO (2)



Lightweight Waterproof Backpack
Short-distance Travel Backpack

BSAM Checker v0.2.1 - Bluetooth Security Assessment Methodology - Tarlogic Security

BSAM Checker

- Audited devices

Audited devices

AUDIT

Brand	Model	Type	Comments	
JBL	Tune Flex	Earbuds		DETAIL

Page 1 of 1 (1 items) 1 10 items per page

Settings

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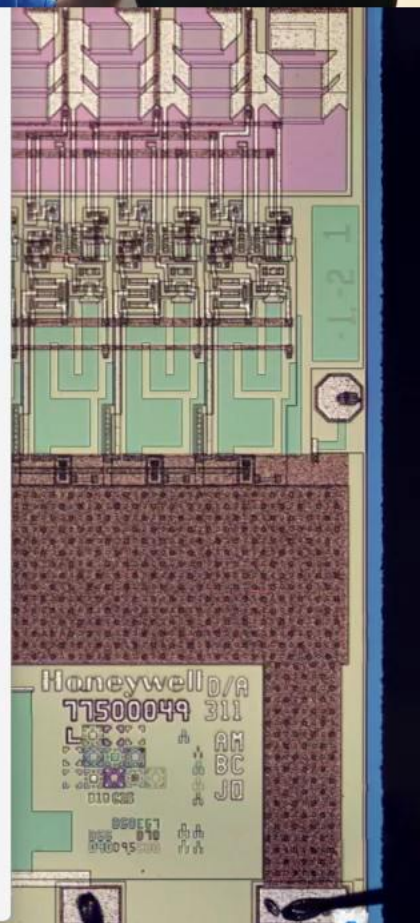
BSAM Checker v0.2.1 - Tarlogic Security © 2026



Papelera de reciclaje

BSAM Checker v0.2.1 - B Windows PowerShell

ESP 17:44 03/03/2026

DEMO (2)

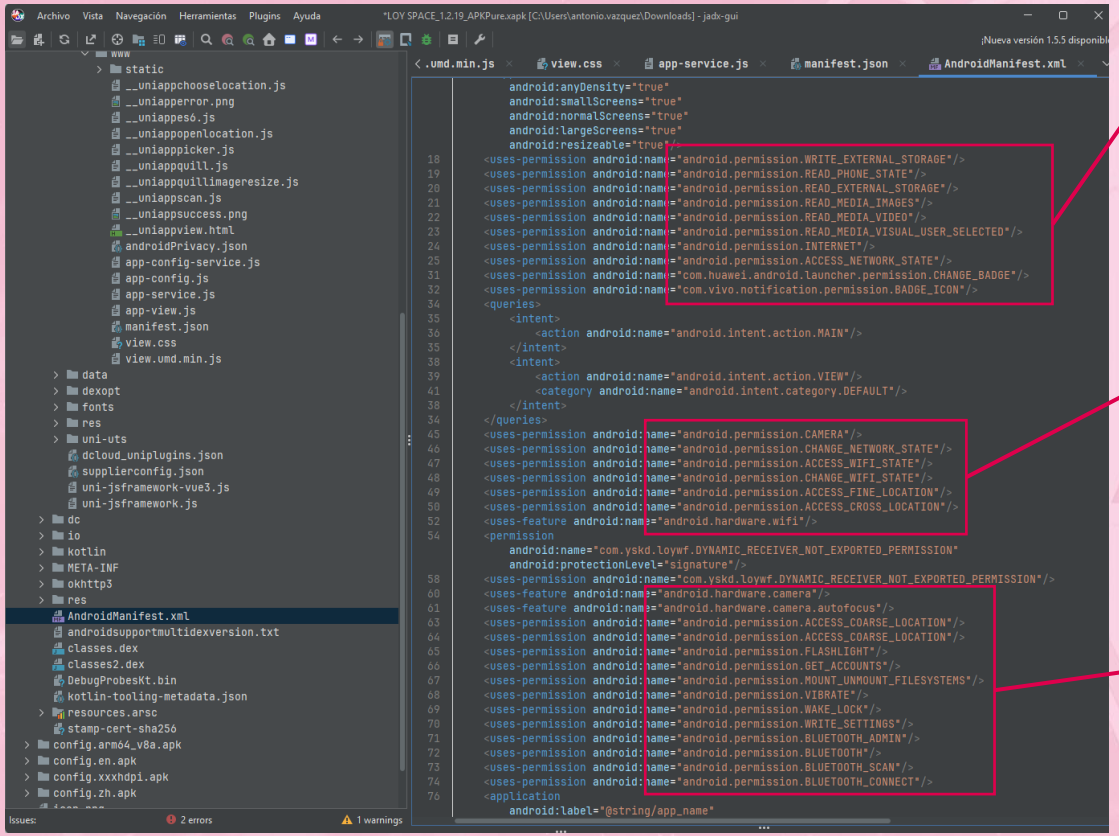
1

- The controls in the **“Bluetooth Service Security”** section are vulnerable…

2

- There is no type of Access control for any service!

DEMO (2)




```
"android.permission.WRITE_EXTERNAL_STORAGE"/>
"android.permission.READ_PHONE_STATE"/>
"android.permission.READ_EXTERNAL_STORAGE"/>
"android.permission.READ_MEDIA_IMAGES"/>
"android.permission.READ_MEDIA_VIDEO"/>
"android.permission.READ_MEDIA_VISUAL_USER_SELECTED"/>
"android.permission.INTERNET"/>
"android.permission.ACCESS_NETWORK_STATE"/>
"com.huawei.android.launcher.permission.CHANGE_BADGE"/>
"com.vivo.notification.permission.BADGE_ICON"/>
```

```
me="android.permission.CAMERA"/>
me="android.permission.CHANGE_NETWORK_STATE"/>
me="android.permission.ACCESS_WIFI_STATE"/>
me="android.permission.CHANGE_WIFI_STATE"/>
me="android.permission.ACCESS_FINE_LOCATION"/>
me="android.permission.ACCESS_CROSS_LOCATION"/>
"android.hardware.wifi"/>
```

```
android.hardware.camera"/>
android.hardware.camera.autofocus"/>
e="android.permission.ACCESS_COARSE_LOCATION"/>
e="android.permission.ACCESS_COARSE_LOCATION"/>
e="android.permission.FLASHLIGHT"/>
e="android.permission.GET_ACCOUNTS"/>
e="android.permission.MOUNT_UNMOUNT_FILESYSTEMS"/>
e="android.permission.VIBRATE"/>
e="android.permission.WAKE_LOCK"/>
e="android.permission.WRITE_SETTINGS"/>
e="android.permission.BLUETOOTH_ADMIN"/>
e="android.permission.BLUETOOTH"/>
e="android.permission.BLUETOOTH_SCAN"/>
e="android.permission.BLUETOOTH_CONNECT"/>
```

DEMO (2)

```
if (filterArray.indexOf("brand") != -1) { getDeviceInfoResult.setBrand(Build.MANUFACTURER); }
if (filterArray.indexOf("deviceBrand") != -1) { getDeviceInfoResult.setDeviceBrand(Build.MANUFACTURER);}
if (filterArray.indexOf("model") != -1) { getDeviceInfoResult.setModel(Build.MODEL); }
if (filterArray.indexOf("deviceModel") != -1) { getDeviceInfoResult.setDeviceModel(Build.MODEL); }
if (filterArray.indexOf("isRoot") != -1) { getDeviceInfoResult.setRoot(Boolean.valueOf(DeviceUtil.INSTANCE.hasRootPrivilege()));
if (filterArray.indexOf("isSimulator") != -1) { getDeviceInfoResult.setSimulator(Boolean.valueOf(DeviceUtil.INSTANCE.isSimulator(uniActivity, false)));
if (filterArray.indexOf("isUSBDebugging") != -1) { getDeviceInfoResult.setUSBDebugging(Boolean.valueOf(DeviceUtil.INSTANCE.listeningForADB()));
```

```
map.put("carrier", Integer.valueOf(carrier));
map.put("cid", Integer.valueOf(baseStationId));
map.put("lac", Integer.valueOf(networkId));
jSONObject.put("type", getNetworkConectivityType(context));
jSONObject.put("mcc", mcc);
jSONObject.put("mnc", mnc);
```

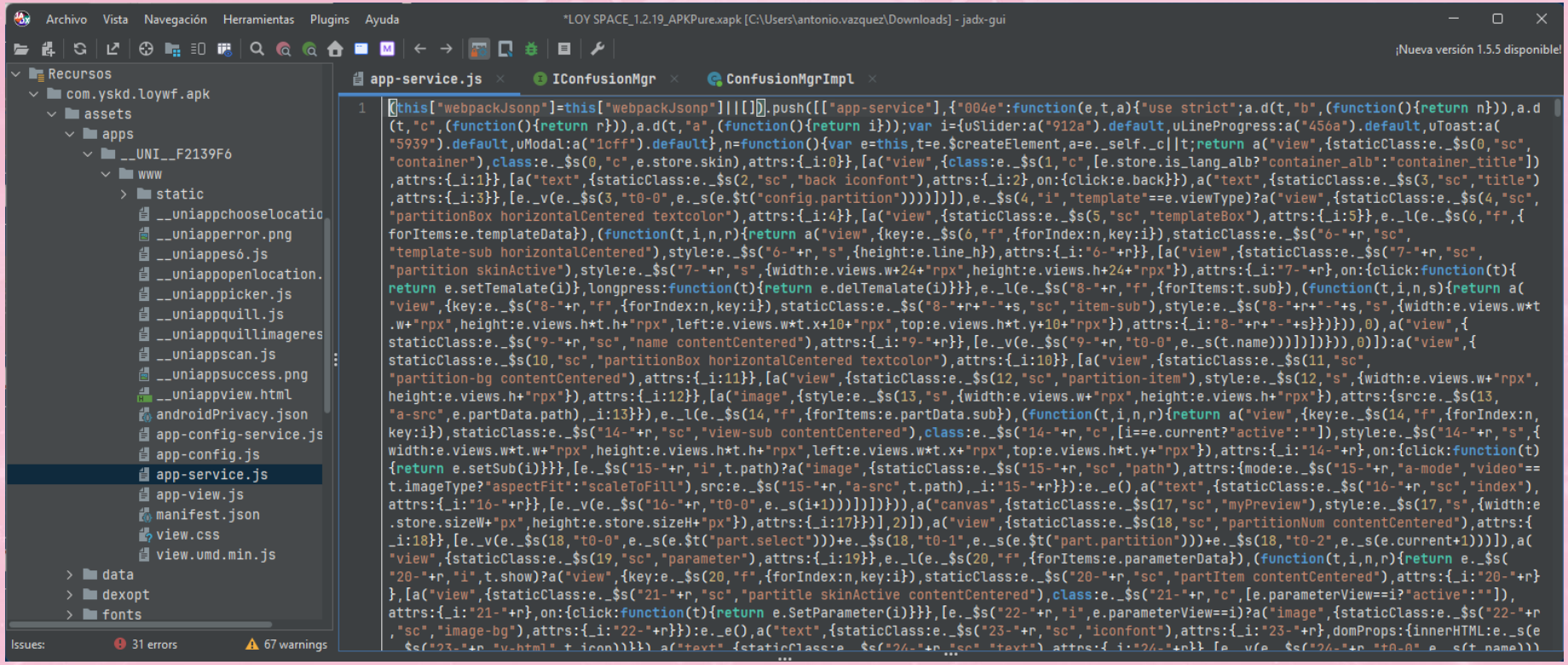
```
String[] imei = AbstractC1915x1.getImei(context);
jSONObject.put("imei", (imei == null || imei.length <= 0) ? "" : TextUtils.join(",", imei));
```

```
jSONObject.put("imsi", imsi);
```

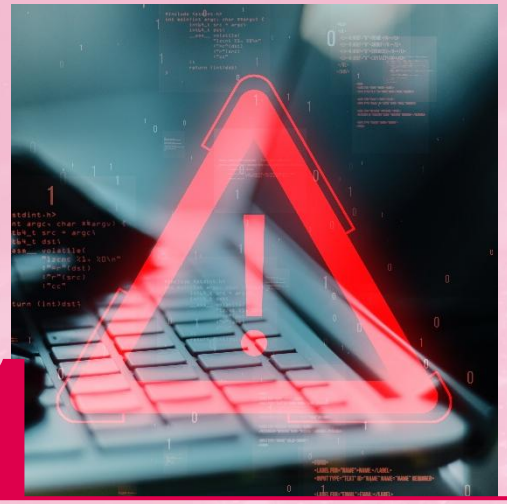
DEMO (2)

```
    public static void setFileAccess(Object obj, IApp iApp, boolean z) {
        if (obj == null || DeviceInfo.sDeviceSdkVer <= 16) {
            return;
        }
        try {
            Class[] clsArr = {Boolean.TYPE};
            Object[] objArr = {Boolean.valueOf(z)}; // setAllowUniversalAccessFromFileURLs, setAllowFileAccessFromFileURLs, setAllowFileAccess
            String strDecodeString = iApp != null ? iApp.getConfusionMgr().decodeString(
"eW9+S2ZmZX1fZGN8b3h5a2ZLaWlveXlMeGVnTGNmb19YRnkqNmEzZDg4ZmEtNGJhMC00NzlmLTk0MjItZTVhYWJlMTU4OTdiNzQ=", true, 10) : AbstractC1794d1.
decodeString("eW9+S2ZmZX1fZGN8b3h5a2ZLaWlveXlMeGVnTGNmb19YRnkqNmEzZDg4ZmEtNGJhMC00NzlmLTk0MjItZTVhYWJlMTU4OTdiNzQ=", true, 10);
            String strDecodeString2 = iApp != null ? iApp.getConfusionMgr().decodeString(
"eG5/SmdnZHxNYmduSmhobnh4TXlkZk1iZ25eWUd4KjZhM2Q4OGZhLTRiYTAtNDc5Zi05NDIyLWU1YWFiZTE1ODk3Yjc1", true, 11) : AbstractC1794d1.decodeString(
"eG5/SmdnZHxNYmduSmhobnh4TXlkZk1iZ25eWUd4KjZhM2Q4OGZhLTRiYTAtNDc5Zi05NDIyLWU1YWFiZTE1ODk3Yjc1", true, 11);
            String strDecodeString3 = iApp != null ? iApp.getConfusionMgr().decodeString(
"f2l4TWBgY3tKZWBpTW9vaX9/KjZhM2Q4OGZhLTRiYTAtNDc5Zi05NDIyLWU1YWFiZTE1ODk3Yjc2", true, 12) : AbstractC1794d1.decodeString(
"f2l4TWBgY3tKZWBpTW9vaX9/KjZhM2Q4OGZhLTRiYTAtNDc5Zi05NDIyLWU1YWFiZTE1ODk3Yjc2", true, 12);
            if (isAllowFileAccessFromFileURLs || !z) {
                PlatformUtil.invokeMethod(obj, strDecodeString, clsArr, objArr);
                PlatformUtil.invokeMethod(obj, strDecodeString2, clsArr, objArr);
            }
            PlatformUtil.invokeMethod(obj, strDecodeString3, clsArr, objArr);
        } catch (Exception unused) {
        }
    }
```

DEMO (2)


Archivo Vista Navegación Herramientas Plugins Ayuda
*LOY SPACE_1.2.19_APKPure.xapk [C:\Users\antonio.vazquez\Downloads] - jadx-gui
¡Nueva versión 1.5.5 disponible!
Recursos
com.yskd.loywf.apk
assets
apps
__UNI__F2139F6
www
static
__uniappchooselocatio
__uniapperror.png
__uniappes6.js
__uniappopenlocation.
__uniapppicker.js
__uniappquill.js
__uniappquillimageres
__uniappscan.js
__uniappsuccess.png
__uniappview.html
androidPrivacy.json
app-config-service.js
app-config.js
app-service.js
app-view.js
manifest.json
view.css
view.umd.min.js
data
dexopt
fonts
app-service.js
IConfusionMgr
ConfusionMgrImpl
Issues:
31 errors
67 warnings

DEMO (1)



1

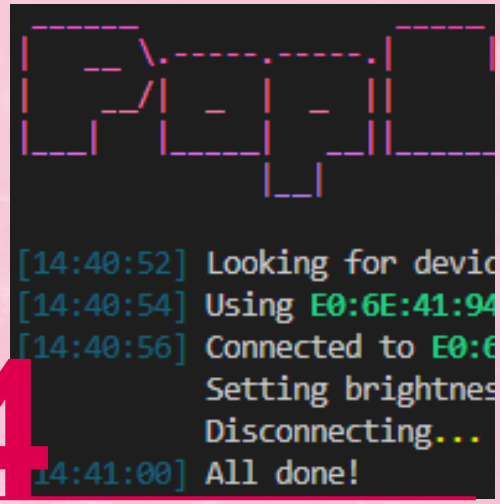
Service-related controls fail…

2

No type of Access control in any of the services!

3

We have reversed the app and created a PoC



4

https://github.com/TarlogicSecurity/BluetoothExamplesAndDemos

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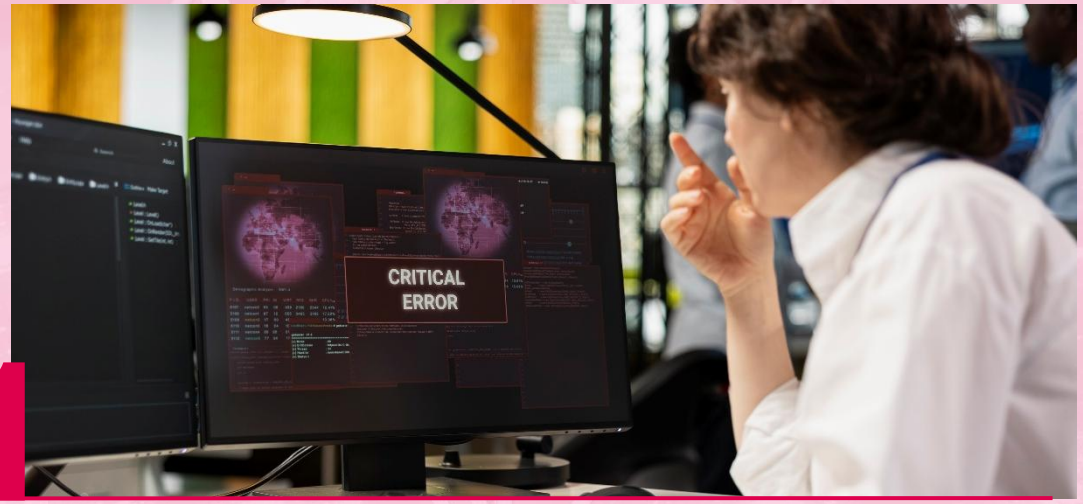
05 LIMITATIONS

LIMITATIONS



1

Not all controls can be automated…

LIMITATIONS

2

Every Bluetooth device is different:

- Many different implementations of the stack
- Unpredictable behaviors
- Free or lax implementations of the standard 2

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06 CONCLUSIONS & FUTURE WORK

CONCLUSIONS

1

Work is ongoing to assist in the audit of the remaining controls

LIMITATIONS

2

Multiplatform audits:

- Mac OS
- Linux

CONCLUSIONS

3

THERE ARE NO EXCUSES NOW…
https://bsamchecker.tarlogic.com/

TARLOGIC
CYBERSECURITY EXPERTS

THANKS

contact@tarlogic.com

www.tarlogic.com
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